

DSA STUDY BLUEPRINT SERIES

95. Kth Smallest in BST

Finding Kth Smallest Element in BSTs

Section 09 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Inorder:** Inorder values increase sorted.
- Traverse inorder, decrementing K; target element found when $K == 0$.

Memory & Execution Blueprint

BST Inorder: [1, 2, 3, 4] $\Rightarrow$ Kth Smallest ($K=2$) is **2**

C++ Implementation Reference

Standard code layout and syntax

```
int ans = -1;
void inorderK(TreeNode* root, int& k) {
    if (!root) return;
    inorderK(root->left, k);
    k--;
    if (k == 0) { ans = root->val; return; }
    inorderK(root->right, k);
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	$O(N)$ / $O(H + K)$ average

Key Practices & Gotchas

- **Important:** Pass count variable by reference to track updates across stack frames.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace BST constraint validation checks verifying node ranges `[Min, Max]`:

Active Node	Valid range limit	Comparison check	Recursive branch step
15 (Root)	$[-\infty, +\infty]$	True $(-\infty < 15 < +\infty)$	Recurse Left child with range $[-\infty, 15]$
10 (Left)	$[-\infty, 15]$	True $(-\infty < 10 < 15)$	Recurse Right child with range $[10, 15]$

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Inorder property:** BST Inorder traversals yield strictly increasing sorted lists.
- **Tree Balancing:** Balanced tree heights (AVL/Red-black) guarantee logarithmic search times.
- **Related LeetCode Problems:** #98 Validate Binary Search Tree, #700 Search in a Binary Search Tree, #108 Convert Sorted Array to Binary Search Tree.